Spending a Fixed Compute Budget: Agents, Rounds, Reasoning Depth, and Sampling

Study `openrouter_agent_allocation_v1`

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Abstract

A fixed inference budget can be spent in several ways. It can buy more agents working in parallel, more rounds of revision, a longer reasoning allowance inside a single call, or simply more independent samples read out by a vote. This report asks which of those purchases is worth making, holding total compute exactly constant.

We define the budget as $C = B \cdot R \cdot A$, the product of the number of agents, the number of rounds, and the per-agent per-round reasoning allowance. The study evaluated 95 configurations on one fixed 400-item panel spanning four benchmarks, with 67,514 graded generations and a 25-sample independent bank per item that makes oracle pass@k available at every budget. Because C is materialised per configuration, allocations that differ in shape but agree in cost can be compared directly. At $C = 8A$, for instance, seven distinct allocations were run.

Three findings hold across benchmarks. First, at matched compute the shape of the allocation barely matters: of 70 per-benchmark comparisons against independent sampling at the same budget, none survives multiplicity correction, and the largest effect anywhere is 6 accuracy points. Second, every allocation plateaus near what two samples and a vote already deliver, while oracle pass@k continues to climb, so the headroom that compute makes available but no configuration collects grows from zero to 6.7 accuracy points on the three-benchmark macro and to 9 points on MMLU-Pro. Third, letting agents see each other's work actively destroys accuracy, and the damage grows with rounds: at five agents and four peer rounds the penalty is 5.0 points on the macro and 10 points on MBPP+.

The binding constraint in this regime is not generation but selection. Extra compute reliably produces a correct answer somewhere in the pool; no evaluated policy reliably finds it.

1 Methods

1.1 Model, benchmarks and panels

Generations come from `google/gemini-2.5-flash-lite` through OpenRouter with the provider pinned, fallbacks disabled and caching off, at temperature 0.7 with nucleus sampling at 0.95. A replication arm uses `google/gemini-2.5-flash`.

Four benchmark families are scored: MATH-500 with `math_verify`, MMLU-Pro with exact option match, a four-subtask zero-shot BBH subset (logical deduction, shuffled objects, boolean expressions, tem-

poral sequences) with exact task match, and MBPP+ graded by the full EvalPlus suite. BBH is reported as this four-subtask subset, not the official full BBH score, and its aggregate weights the four subtasks equally.

The main panel is 400 items, 100 per benchmark. A high-allowance panel of 96 items and a replication panel of 48 items are nested subsets of it, so every cross-panel comparison uses shared item ids only. All 95 main-panel configurations share one fixed item population, which is what allows every contrast in this report to be paired item by item.

1.2 The compute accounting

Each configuration is a triple (B, R, A) and the budget is their product. The sealed tables carry B, R and A as explicit columns, so no parsing of configuration names is involved:

Family	Notation	B	R	C
Independent samples, vote	`sample/k`	k	1	kA_0
One agent revising	`private_single/t`	1	t	tA_0
Team, no peer exposure	`private_team/n/r`	n	$1+r$	$n(1+r)A_0$
Team, peer exposure	`peer/n/r`	n	$1+r$	$n(1+r)A_0$
One deeper call	`deep/m`	1	1	mA_0

A_0 is 1024 tokens on the main panel and 4096 on the high panel. The `deep` family scales A rather than the call count, so its budget must be read from A and not from the number of generations; a build-time check asserts that the derived C equals the sealed requested reasoning tokens for every configuration on every panel, and that $B \cdot R$ equals the sealed generation count everywhere except `deep`, where it is one call by construction.

Figure 1 shows the resulting design. Fourteen budget levels carry two or more distinct allocations, and the levels $C/A_0 \in \{4, 8\}$ carry seven each.

1.3 What counts as a result, and what does not

A configuration delivers exactly one answer per item. Text benchmarks select it by plurality vote; MBPP+ selects by prompt-visible public tests only, and is therefore reported as `public_select@k` and never as a vote. Delivered accuracy counts operational failures, unparseable outputs and abstentions as incorrect, so nothing is silently dropped.

Oracle pass@k is not an available selector. It is reported as a ceiling only, using the standard combinatorial estimator $1 - \binom{n-c}{k} / \binom{n}{k}$ over the 25-sample bank. Two normalised quantities are used throughout:

Amplification is delivered accuracy minus the accuracy of independent sampling at the same budget, in accuracy points, paired per item. It answers whether an allocation beats simply drawing $k = C/A_0$ samples and reading them out.

Ceiling capture is $\eta = (\text{accuracy} - \text{pass@1})/(\text{coverage@}k - \text{pass@1})$, the share of the headroom that the budget makes available which the configuration actually delivers. The denominator is the realised prefix coverage of the same k samples, which is binary per item and therefore pairs with the numerator under the same resample indices. A value of 1 means the configuration collected everything the budget put within

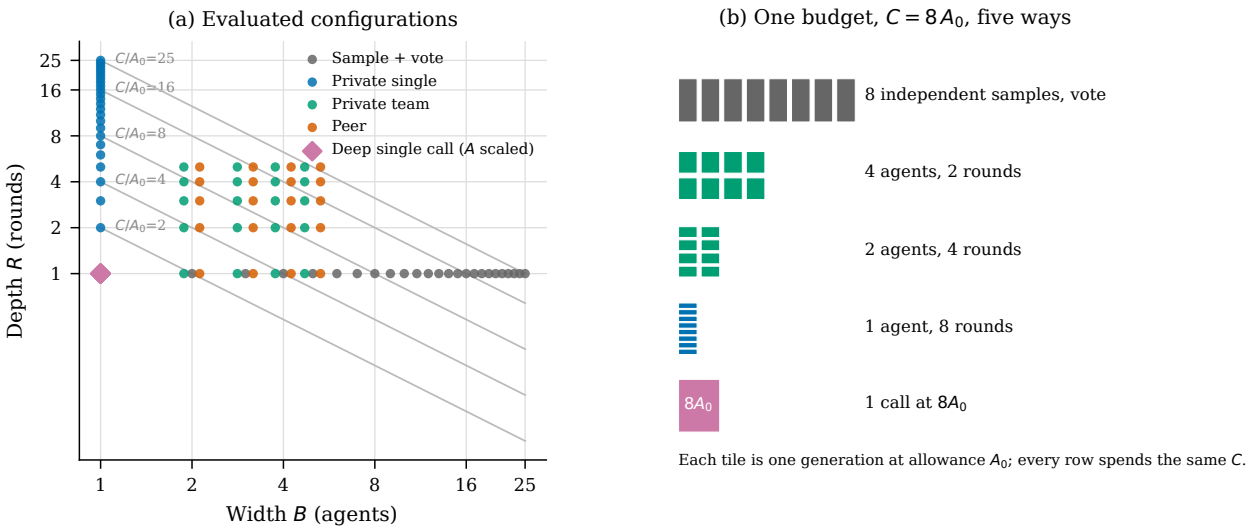


Figure 1: The design. (a) Every evaluated configuration in the width-depth plane; grey lines are iso-compute contours, along which every point spends the same C. Private-team and peer points are nudged apart because they occupy identical (B, R) cells and differ only in whether agents see each other. (b) The five ways one budget of $8A_0$ was actually spent.

reach; 0 means it did no better than one sample. Values above 1 are possible and meaningful, because a revising or communicating configuration can produce an answer that independent sampling never generated. We suppress η where the denominator falls below 0.02, which masks BBH entirely.

1.4 Estimation and multiplicity

All intervals are percentile intervals from a 10,000-replicate item-clustered bootstrap, stratified by benchmark and with BBH subtasks weighted equally. Every configuration shares the same resample indices, so differences between configurations are correctly paired.

Two contrasts were prespecified and frozen before collection: peer versus private team at $B{=}4$, $r{=}4$, and the communicate-then-private decomposition at $h{=}3$. Those carry Holm-corrected p-values and are quoted directly from the sealed contrast table. Everything else in this report is exploratory. In the amplification matrix we mark cells whose nominal paired interval excludes zero, and we also report how many survive Benjamini-Hochberg control within each panel, which turns out to be the more informative number.

Macro averages are reported two ways: over all four benchmarks, which reconciles with the run report, and over the three informative ones. BBH is saturated at 1.000 in every configuration, so including it inflates the level and compresses every contrast toward zero.

1.5 Limitations

A is a requested allowance, not realised reasoning; realised token counts are recorded separately and are consistently below the request. This is a token-accounted and dollar-accounted study, not a FLOP-matched one. With 100 items per benchmark, one item is one accuracy point, so differences below roughly 3 points are not separable from noise. Results are from a single small model, and the peer-communication finding in particular should not be assumed to transfer to larger models. The text completion headroom was amended

twice during piloting, from 1024 to 2048 to 4096 tokens, and the main run was collected at 4096 under a clarified system prompt. Frontiers shown here are descriptive: the best among evaluated configurations on this sample, never a validated policy.

2 Results

2.1 Allocation shape barely matters; the ceiling keeps rising

Figure 2 is the headline. On every benchmark the oracle ceiling climbs steadily with compute while every delivered curve flattens almost immediately.

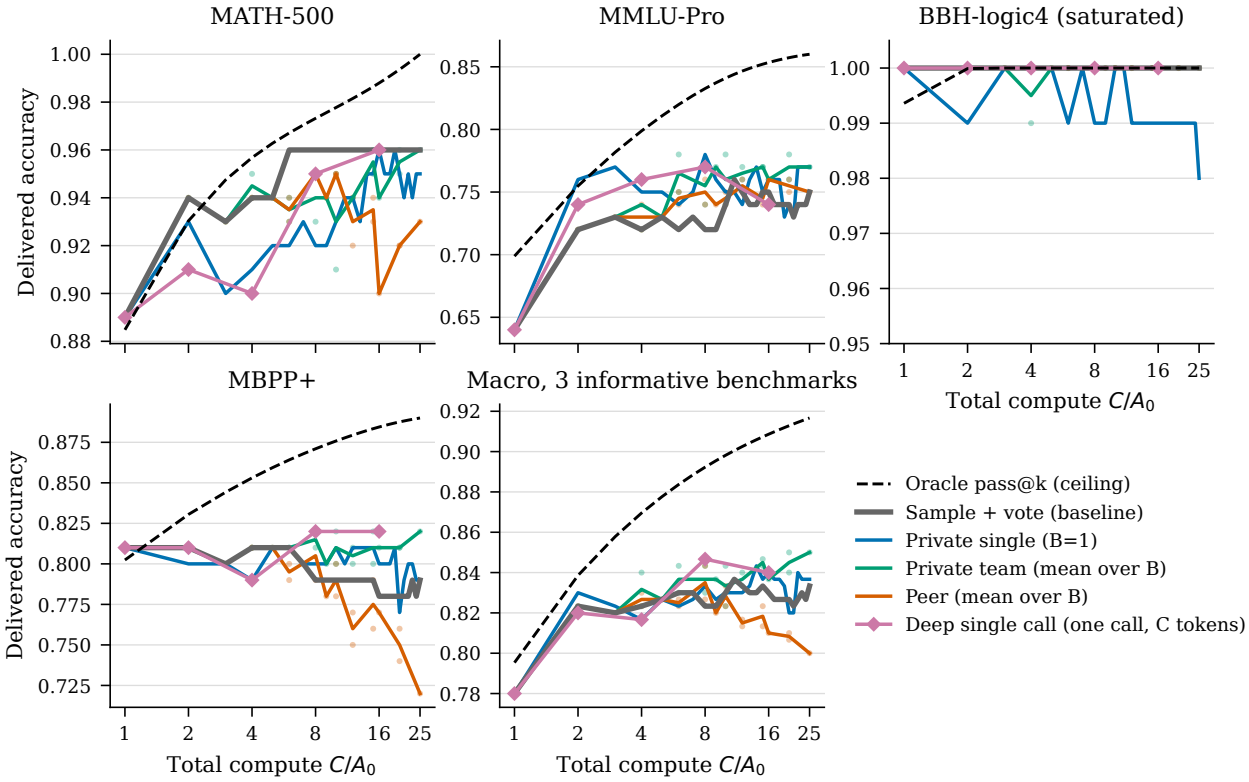


Figure 2: Delivered accuracy against total compute. All curves at a given x spend the same budget. The dashed ceiling is oracle pass@k, an unavailable selector. Faint points are individual team sizes; family lines are the mean over B. BBH is drawn on a fixed window because it is saturated.

On the three-benchmark macro a single sample delivers 0.780 and two samples with a vote deliver 0.823. Twenty-five samples deliver 0.833, an additional 1.0 point for twelve times the compute. The best configuration anywhere in the design is a five-agent private team with three revision rounds at 0.850, which is 2.7 points above sample-and-vote at a comparable budget and 7.0 points above one sample. Over the same range oracle pass@k rises from 0.780 to 0.917.

Per benchmark the picture differs in an instructive way. On MATH-500 plain sampling is essentially unbeatable: it reaches 0.960, which is the best value any configuration attains, while the oracle reaches 1.000. On MMLU-Pro allocation helps most, with a three-agent private team at one revision round reaching 0.780 against 0.720 for two samples. On MBPP+ the achievable readout actually *declines* with budget, from 0.810

at one sample to 0.790 at twenty-five, while oracle coverage rises to 0.890: selecting code by prompt-visible tests gets worse as the candidate pool grows, because a candidate that passes the visible assertion and fails the hidden suite becomes easier to find.

Benchmark	pass@1	vote@2	vote@25	best any	oracle@25	headroom
MATH-500	0.890	0.940	0.960	0.960	1.000	0.040
MMLU-Pro	0.640	0.720	0.750	0.780	0.860	0.090
BBH-logic4	1.000	1.000	1.000	1.000	1.000	0.000
MBPP+	0.810	0.810	0.790	0.820	0.890	0.070
Macro (3)	0.780	0.823	0.833	0.850	0.917	0.067

2.2 The amplification matrix

Figure 3 places every allocation against independent sampling at the same budget.

The matrix is mostly pale, and that is the result. Across 70 comparisons per benchmark, zero survive multiplicity correction; the smallest p-value on the macro is 0.017, far above the Benjamini-Hochberg threshold. Only 2 macro cells and 8 MATH-500 cells clear even an uncorrected interval. A practitioner choosing among these allocations at fixed cost is, on this evidence, choosing among options that cannot be reliably distinguished.

The signs are nonetheless consistent and worth reading. Private teams sit on the positive side on MMLU-Pro (up to $+6$ points) and MBPP+ (up to $+3$), and on the negative side on MATH-500. Peer configurations sit on the negative side almost everywhere and grow more negative toward the bottom of the matrix, that is, at larger budgets with more rounds. The `deep` column is positive on MBPP+ ($+3$ to $+4$) and negative on MATH-500.

Figure 4 normalises the same comparison against the headroom available at each budget.

Capture falls as budget rises. On the macro the median η is 0.67 at the smallest budgets and 0.39 by $C/A_0 \geq 16$; on MMLU-Pro it falls from 0.80 to 0.50. On MBPP+ the median capture at large budgets is -0.25, meaning the typical configuration there ends up below a single sample even as coverage improves. The one cell above 1.0, a single agent revising once on MMLU-Pro, is genuine: revision produced correct answers that neither independent sample contained.

2.3 Width against depth

Figure 5 isolates the shape question by fixing the budget and varying only how it is split.

For private teams, wider is mildly better than deeper: on the macro the mean amplification rises from about 0 at $B{=}1$ to $+1.6$ points at $B{=}5$, and the same upward slope appears on MATH-500, MMLU-Pro and MBPP+. Pure depth, one agent revising many times, is the weakest private allocation on every informative benchmark. For peer configurations the slope is absent or reversed and the whole curve sits below zero on the macro and on MBPP+.

Two readings follow. Spreading a budget across a few agents is slightly better than spending it on one agent thinking for longer in sequence, but the advantage is within noise. And the gap between the private and peer curves, which is the pure effect of letting agents see each other, is larger and more consistent than the

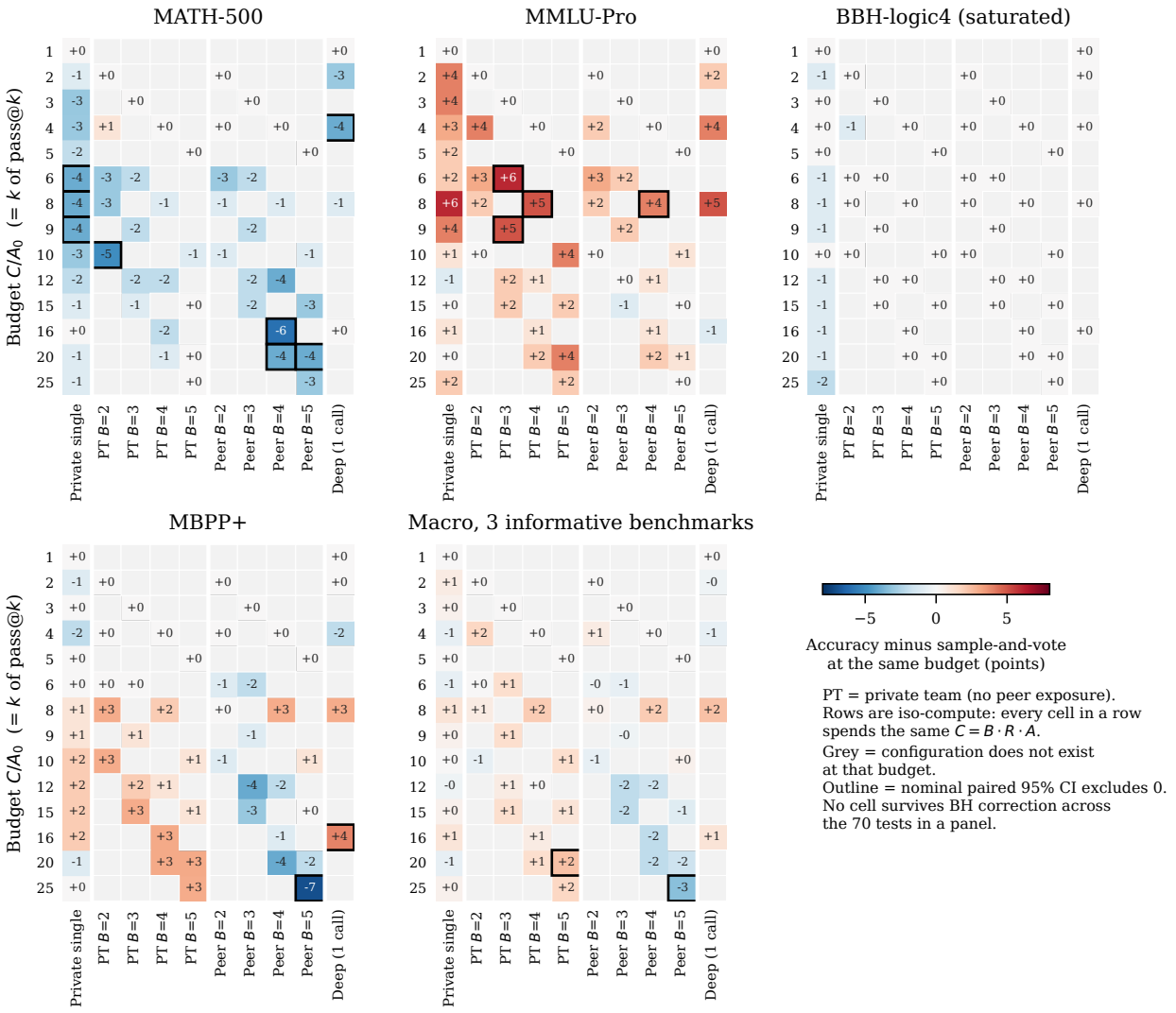


Figure 3: Amplification relative to sampling. Rows are iso-compute budgets; every cell in a row spends the same C. Values are delivered accuracy minus sample-and-vote at that budget, in accuracy points. Outlines mark nominal paired intervals excluding zero; *no* cell survives Benjamini-Hochberg control across the 70 tests in a panel. PT is a private team, that is, a team without peer exposure.

width-depth effect it is plotted against.

2.4 Generation is not the bottleneck

Figure 6 makes the central claim directly.

At the largest budget the best configuration on the macro reaches 0.850 against an oracle of 0.917, leaving 6.7 points unrealised. Per benchmark the unrealised headroom is 4.0 points on MATH-500, 9.0 on MMLU-Pro and 7.0 on MBPP+. The band widens monotonically: at $C = 2A_0$ it is near zero, and every additional unit of compute after that adds more to the ceiling than to what any policy delivers.

The blue line is deliberately generous. It takes the maximum over up to seven configurations at each budget, so it is biased upward by selection on this very sample, and the gap is still large.

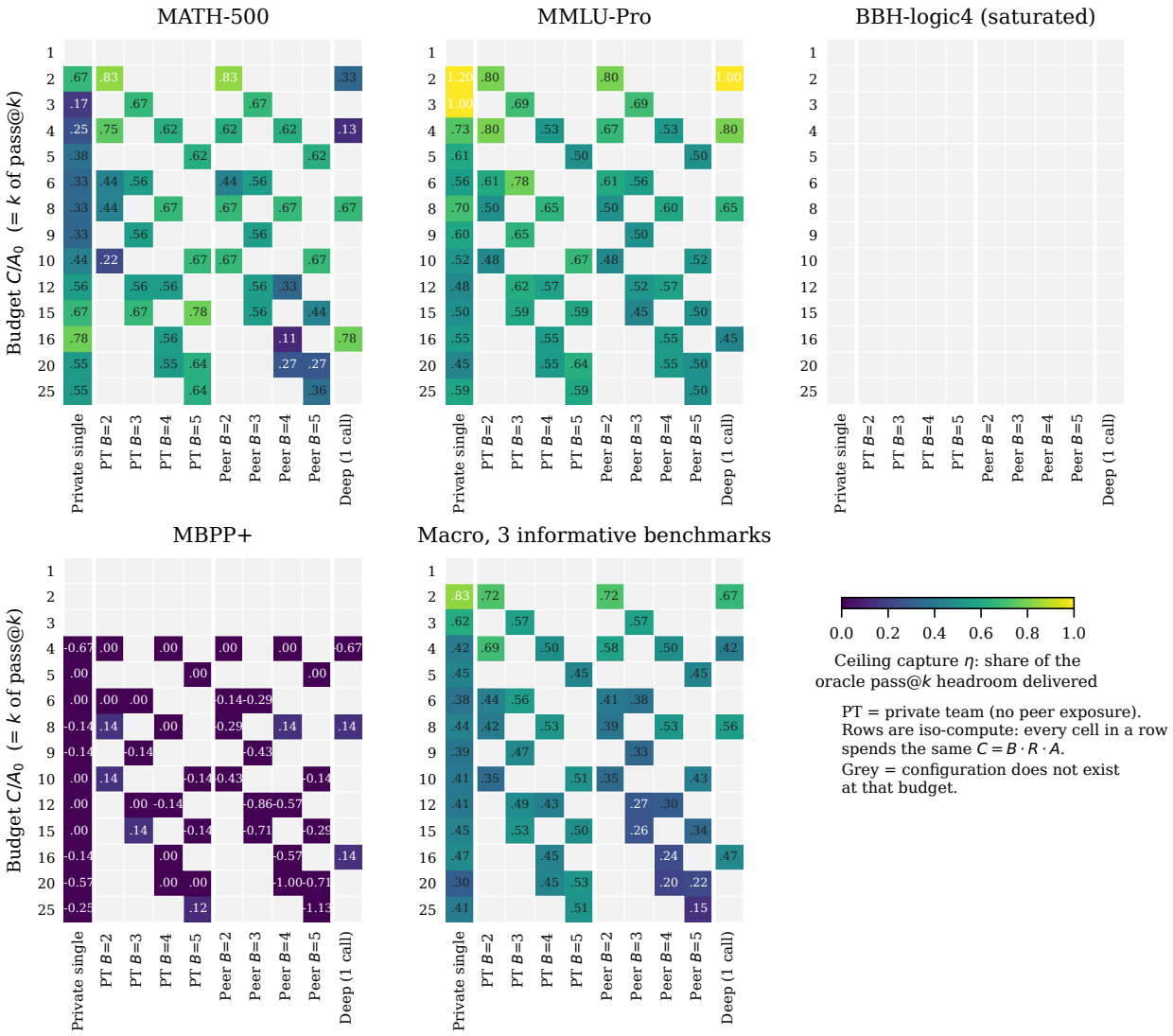


Figure 4: Ceiling capture η: the share of the headroom a budget makes available that the configuration actually delivers. BBH is masked because its denominator is zero.

2.5 The cost of letting agents talk

The one prespecified contrast that resolves is the comparison between teams that see each other and teams that do not, at matched size and matched rounds.

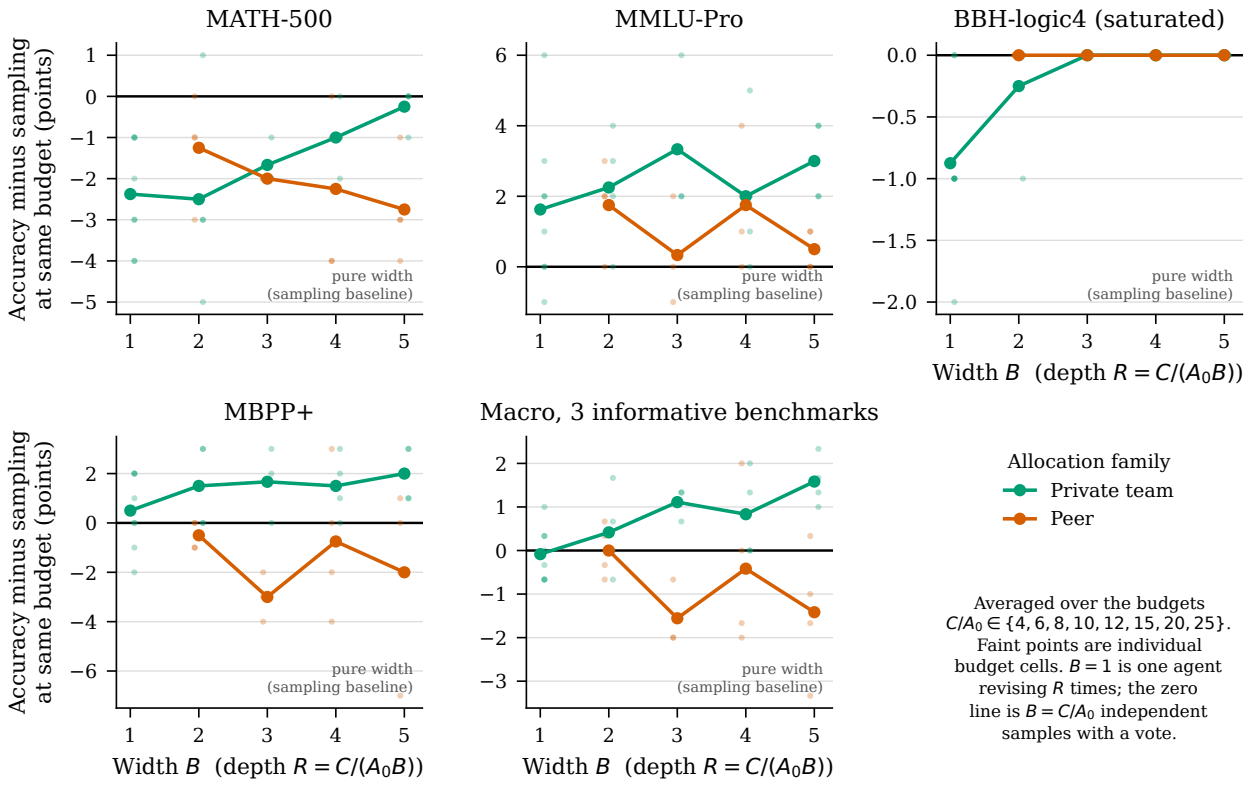


Figure 5: Allocation shape at fixed compute, averaged over budgets $C/A_0 \in \{4, 6, 8, 10, 12, 15, 20, 25\}$. B=1 is one agent revising R times. The zero line is the same budget spent on $B = C/A_0$ independent samples.

Contrast	Level	n	Estimate	95% CI	Holm p
Peer − private team, B=4, r=4	Macro (4)	400	-0.0250	[-0.046, -0.008]	0.027
	MATH-500	100	-0.0300	[-0.064, -0.003]	n/a
	MMLU-Pro	100	+0.0000	[+0.000, +0.000]	n/a
	BBH-logic4	100	+0.0000	[+0.000, +0.000]	n/a
	MBPP+	100	-0.0700	[-0.144, -0.010]	n/a
CP − private-team gain, h=3	Macro (4)	192	-0.0052	[-0.025, +0.009]	0.567
	MATH-500	48	-0.0208	[-0.021, +0.004]	n/a
	MMLU-Pro	48	+0.0000	[+0.000, +0.000]	n/a
	BBH-logic4	48	+0.0000	[+0.000, +0.000]	n/a
	MBPP+	48	+0.0000	[-0.053, +0.053]	n/a

Peer exposure at four agents and four rounds costs 2.50 accuracy points on the four-benchmark macro, with a 95% interval of $[-0.046, -0.008]$ and a Holm-corrected p of 0.027. The effect is carried by MBPP+ (-7.0 points) and MATH-500 (-3.0); it is exactly zero on MMLU-Pro and on saturated BBH. The second prespecified contrast, the communicate-then-private decomposition at h=3, is null at -0.005 with $p = 0.567$.

Figure 7 shows that this is not a single unlucky cell but a trend.

The penalty deepens with rounds on every informative benchmark and with team size on MBPP+. At the largest configuration, five agents and four peer rounds, the cost is 5.0 points on the macro ($[-0.077, -0.023]$)

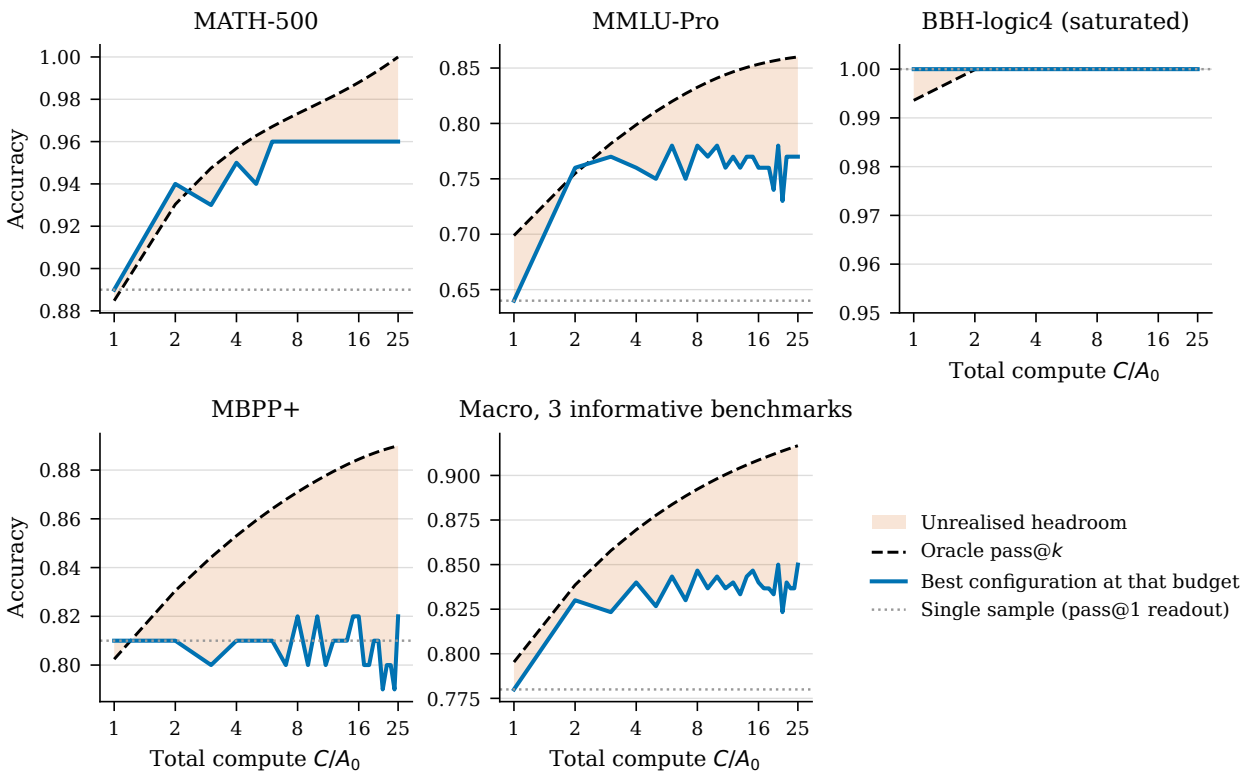


Figure 6: The selection gap. The shaded band is headroom that compute made available and no evaluated configuration collected. The blue line is the maximum over all configurations at that budget, so it is an optimistic envelope and still falls short.

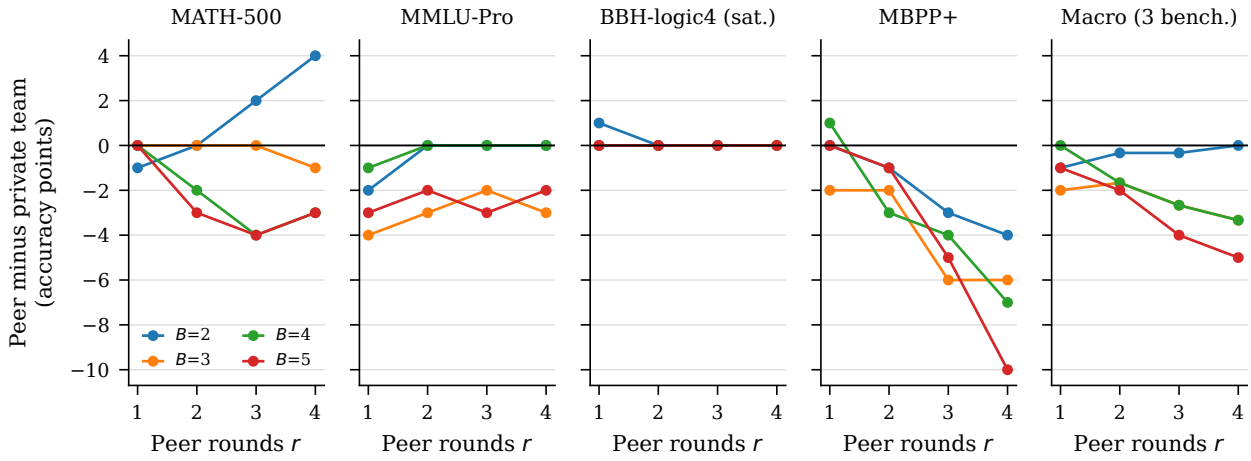


Figure 7: Peer minus private team at matched team size and matched rounds. The only difference between the two arms is whether agents see each other's drafts.

and 10.0 points on MBPP+ ($[-0.17, -0.04]$).

Figure 8 shows the mechanism. Revision rounds move a substantial number of answers in both directions, and the net is close to zero.

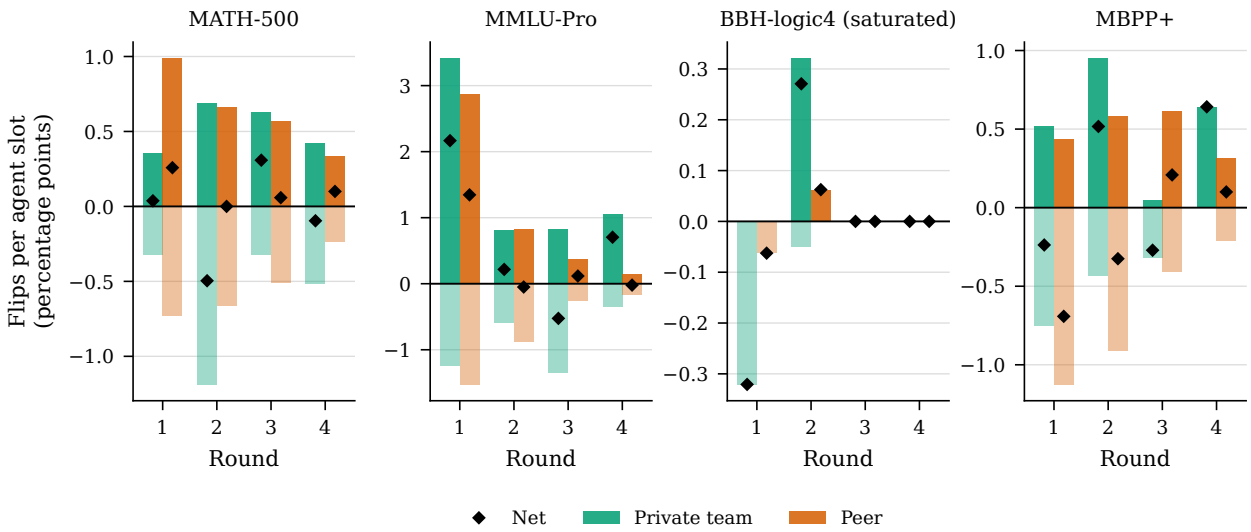


Figure 8: Churn per revision round. Solid bars are wrong-to-correct flips, faded bars are correct-to-wrong, and the diamond is the net. Rates are per agent slot, in percentage points.

On MMLU-Pro the first round has a clearly positive net and later rounds do not, which is why one revision round is the best MMLU-Pro configuration in the whole design. On MATH-500 and MBPP+ the net is negative for several rounds. Peer exposure raises the correct-to-wrong rate more than it raises the wrong-to-correct rate, which is exactly the shape of a consensus process that propagates confident errors.

2.6 Depth inside one call

The `deep` arm spends the whole budget on a single call with a larger reasoning allowance, so it pays no repeated prompt cost and has no opportunity to select.

One deeper call is competitive with everything else and is the single best configuration on MBPP+ (0.820 at $C = 8A_0$, against 0.790 for eight samples). On the macro it reaches 0.847, within 0.3 points of the best configuration in the entire design, using one generation rather than twenty. On MATH-500 it trails sampling slightly. Since it requires no orchestration, no repeated context and no selector, it is the most practical of the allocations tested.

2.7 What this costs in dollars

The frontier saturates almost immediately. One sample costs \$0.000475 per item and one deeper call at $8A_0$ costs \$0.001207; five agents over four peer rounds cost \$0.015196, which is 32 times the single sample, and deliver less. Everything beyond the deeper single call at \$0.001207 per item is, on this evidence, spent without return.

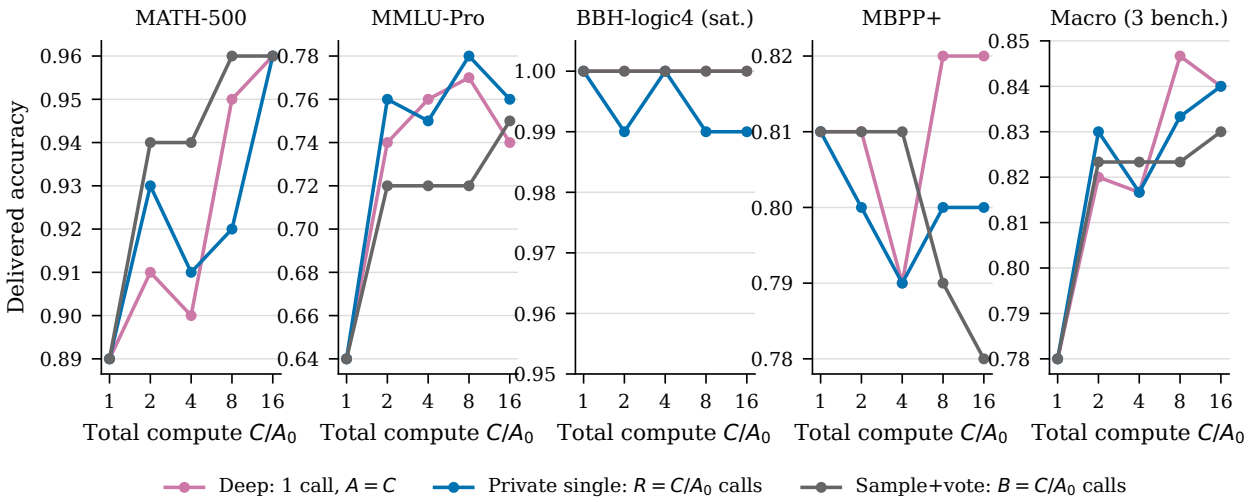


Figure 9: The same budget as one long call, as sequential revisions, or as parallel samples.

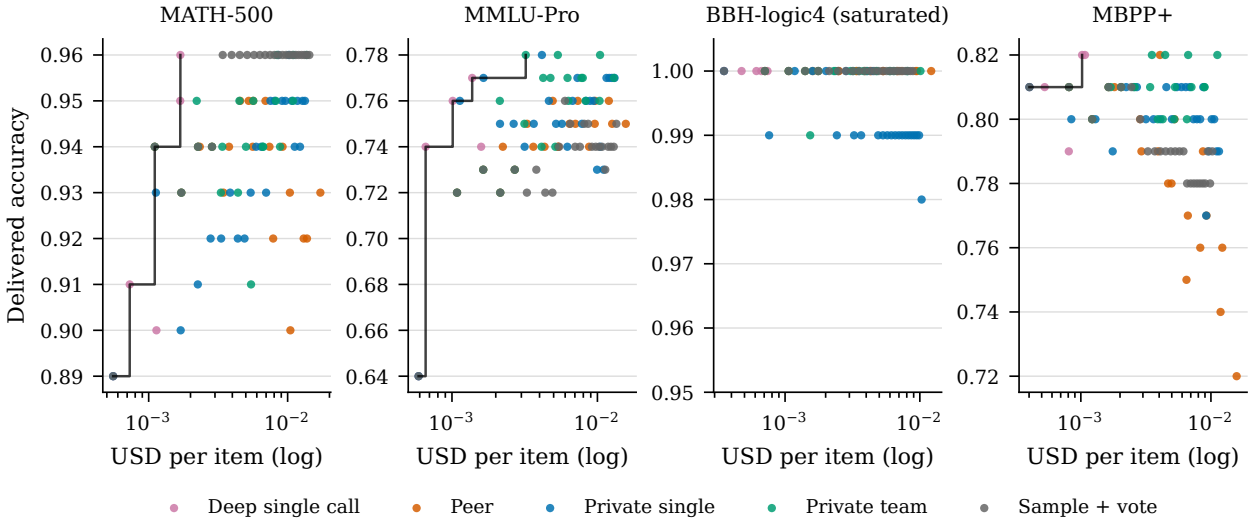


Figure 10: Delivered accuracy against realised spend per item. The step line is the descriptive Pareto front over evaluated configurations.

2.8 Robustness

Raising the allowance to $A_0 = 4096$ lifts the whole picture, with pass@1 at 0.833 and the best configuration reaching 0.903, but the qualitative shape survives: the ceiling is 0.931 and 2.8 points remain unrealised. The Gemini 2.5 Flash replication behaves the same way, with a best configuration of 0.806 against a ceiling of 0.833 at $k = 16$. Both panels are small, 96 and 48 items, so these are consistency checks rather than independent confirmations. The direction of the peer penalty is preserved on both.

2.9 A design check

On all 400 items, a team of n agents run for zero revision rounds delivers exactly the same answer as n independent samples, under both the private and the peer arms. This is true by construction, since with no

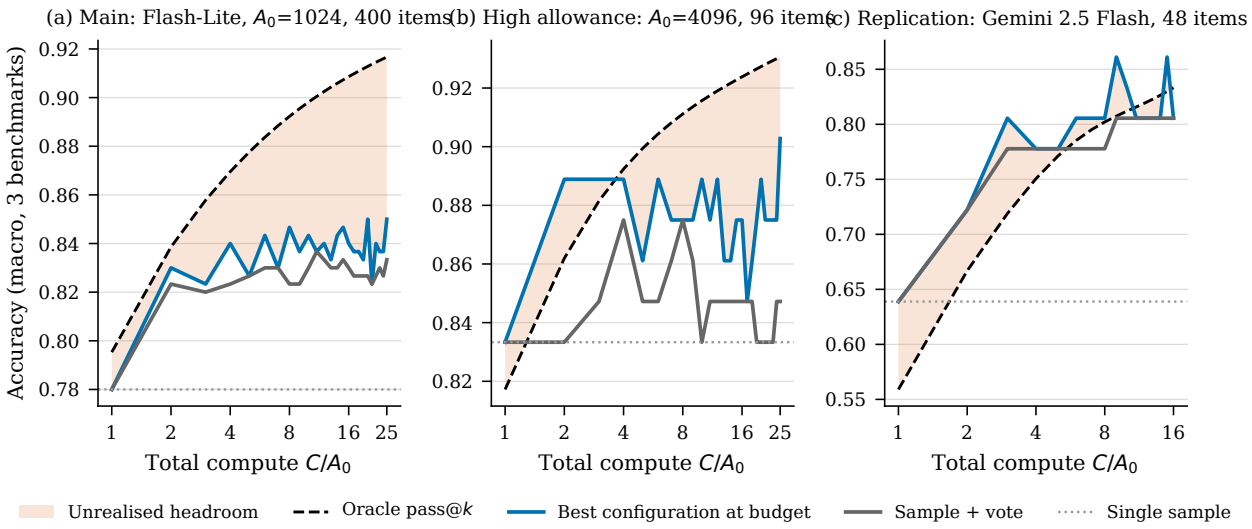


Figure 11: The same comparison at four times the reasoning allowance and on a stronger model, on shared items only. The 96-item and 48-item panels have unstable intervals.

rounds there is nothing to revise and nothing to see, and its exact agreement (400 of 400 items for every n) confirms that the iso-compute re-keying used throughout this report lines up with the sealed selection records.

3 Interpretation

At fixed compute, the shape of the allocation is close to irrelevant. This is the least exciting reading of the data and the best supported one. Seventy paired comparisons per benchmark produce no effect that survives correction, and the spread across all allocations at the largest budget is about 5 accuracy points on the macro, comparable to the width of a single confidence interval. Claims that multi-agent structure beats single-agent inference should be checked against a compute-matched sampling baseline, because on this model and these benchmarks that baseline is already most of the way to the best result anyone achieves.

Test-time compute keeps working; test-time *selection* does not. Oracle pass@k rises steadily and without any sign of saturating below 0.92 on the macro, which means additional compute is still producing correct answers. What fails is the readout. Plurality voting stops improving after roughly two samples, and on MBPP+ the public-test selector gets actively worse as the pool grows, because more candidates means more candidates that satisfy the visible evidence and fail the hidden suite. The gap between what was generated and what was delivered is the largest single quantity in this study, and it grows with every dollar spent.

Communication between agents is a cost, not a benefit, in this regime. The prespecified contrast is negative and significant, the effect deepens monotonically with rounds, and the round dynamics show why: peer exposure increases the rate at which correct answers become wrong more than it increases the reverse. A small model shown its peers' drafts converges, and convergence on a wrong answer is indistinguishable at readout time from convergence on a right one. The private team, which gets the same number of agents and rounds with no mutual visibility, dominates it everywhere. This should be read as a statement about small models with a plurality readout, not about multi-agent debate in general, but it does mean the burden of proof runs the other way.

Width beats depth slightly, and one deeper call beats both on cost. Among private allocations, spreading a budget over three to five agents edges out spending it on one agent revising many times, though the margin is inside the noise. More striking is that a single call with the whole budget as reasoning allowance matches the best orchestrated configuration on the macro to within 0.3 points and wins outright on MBPP+, while requiring no orchestration, no repeated context and no selector. If a practitioner has to pick one thing to do with extra tokens on this model, the evidence points at that.

What would actually move the frontier. Not more agents, and not more rounds. The 6.7-point macro gap is a selection problem, so the levers are selection ones: a verifier that scores candidates on something other than agreement, a code selector that does not overfit to visible tests, or a reranker calibrated against held-out evidence. On MBPP+ the entire 7-point gap is reachable by a selector that avoids being fooled by the public assertion, and the study already shows that the correct program is usually present in the pool. Until the readout improves, buying more generations mostly buys a larger pool of answers nobody can pick from.